BYO AI ACCOUNT BRIDGE / RESEARCH SERIES

MVP & Launch Execution

Milestones, backlog, tests, go/no-go criteria and pilot economics

06

DETAILED TECHNICAL DOSSIER

User-owned accounts / Local-first runtime / Optional cloud relay

Research cutoff: 26 October 2026

A Consumer-web adapter (permission-gated)

B Official account authorization where supported

C User-owned free APIs and locally running models

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> Research notation: [Sxx] markers link to first-party terms, official technical documentation, published standards or original project repositories. Statements labeled proposed, target, or estimate are design judgments, not observed performance.

Execution thesis

Build the smallest developer tool that proves the core experience: one user connects their own permitted AI access on their own machine, calls one unified endpoint, sees an honest source label and can choose a privacy-preserving fallback. The central A-style browser architecture must be demonstrated with a **mock or explicitly approved consumer UI**, not publicly enabled against providers whose terms restrict automation. The product should still give immediate value through official free-tier APIs, Ollama, and any eligible provider-sanctioned account grant. **[S01] [S02] [S04] [S09] [S13]**

Proposed delivery horizon: six to nine weeks for a lean alpha with one full-time engineer and part-time design/security support; this is an estimate, not a commitment. A polished cross-platform beta, provider approvals and external security review could take substantially longer. Every phase has a measurable gate.

Phase map and decision gates

Phase	Duration estimate	Build	Go / no-go criterion
P0 - Legal and feasibility	Week 1	Permission matrix, allowed routes, mock adapter experiment	At least one releasable C path; A consent/terms status documented
P1 - Local foundation	Weeks 2-3	Desktop daemon, keychain, REST, routing, mock A, Ollama	SDK demo works, no unauthenticated local calls
P2 - Useful account access	Weeks 3-5	One free official API, eligible B if authorized, quota/errors	95% goal for eligible controlled test cases
P3 - Remote dashboard	Weeks 5-7	Enrollment, device status, scoped remote keys, WSS relay	Revocation and cross-tenant authorization pass
P4 - Private beta/hardening	Weeks 7-9	Installer, updater, metrics, docs, 10-20 users	Privacy, reliability and usability gates met
P5 - Real A connector	Independent permission timeline	Approved consumer UI adapter	Written provider permission + security/QA sign-off

P5 is not automatically scheduled after P4; it is contingent on provider authorization. Do not make an investor or landing-page claim that P5 will be available on a fixed date.

Week-by-week work breakdown

Week 1: research validation and permission gate

- Create a spreadsheet or issue registry per provider: terms URL/version, automated-access permission, owner, legal finding, eligible regions/accounts and next review date.
- Interview 8-12 budget-conscious developers about current tool, budget, available account, sensitivity of prompts and willingness to keep a local process running.

- Validate three integration targets: OpenAI-compatible SDK caller, terminal curl and one MCP client. Check if the selected B account-plan program admits your distribution model. **[S04] [S07] [S14]**
- Build a tiny fake consumer chat page under your control and test a Playwright adapter that detects loading, final message, timeout and login expiry. **[S21]**
- Output artifacts: provider decision matrix, interview synthesis, two-minute local demo and threat model.

Gate: If no provider allows consumer web automation, retain A as a permission-seeking research module and ship C/B capability with truthful language. If free APIs are also unusable for target geography, re-scope to Ollama-first.

Weeks 2-3: local core and API contract

- Create monorepo `apps/bridge`, `apps/dashboard`, `packages/contracts`, `packages/adapters`, `packages/mcp` (latter dormant until P4).
- Implement `GET /health`, `GET /v1/models`, `POST /v1/chat/completions`, key generation, strict schema validation, cancellation, request IDs, bounded queue and local config.
- Implement `ProviderAdapter` interface and mock browser provider, plus Ollama connector and normalized error mapping.
- Store secrets in OS vault and connection metadata locally; defaults: no prompt logs, key-required localhost, no paid route, no remote relay.
- Add sample `curl`, Node OpenAI SDK and Python OpenAI SDK snippets against local base URL.
- Set up CI: lint, typecheck, unit test, package scan, containerized integration test and signed release pipeline design.

Gate: Two SDKs and curl complete text requests; app rejects invalid local keys, unsupported modalities and unauthorized foreign web Origin; cloud outage has no impact on local mode.

Weeks 3-5: provider reality and graceful failure

- Add one official free-quota provider (Gemini or Groq), with model discovery where safe, documented free-tier caveats and API key held locally. **[S09] [S10] [S11]**
- Add per-route provider request budget, token visibility only where truly provided, measured quota/error classification, exponential backoff and user-owned fallback consent.
- Research and prototype sanctioned OpenAI account grant in eligible local OSS mode; include limits (`store:false`, Responses-only and excluded tools). Do not ship outside approved distribution and hosting scope. **[S04] [S05] [S39]**
- Build connection UX: connect, test, pause, disconnect, delete local secret, see provider's policies and quota status.
- Add snapshot tests for normalized errors, streaming and provider model catalog; exercise API-key revocation and logout.

Gate: A test user can create an API-like request with zero paid-API requirement if they have eligible quota or local hardware; all routing disclosures are understandable and accurate.

Weeks 5-7: cloud dashboard with device-bound relay

- Build email/passkey or reputable managed sign-in; create device registry and dashboard status cards.
- Add one-time device pairing with local confirmation, public-key identity, outbound WSS and heartbeat.
- Add remote gateway keys with user, device, scope, rate ceiling and expiry; implement owner authorization per request.
- Build tunnel reverse streaming with request correlation, limits, cancel and `device_offline` response.

- Cloud holds device metadata and hashed remote keys, not browser profiles/provider secrets; instrument minimal count/duration metrics.
- UI consent explicitly says the remote relay processes prompt/response in transit and computer must stay online.

Gate: Cross-user device request is consistently denied; disconnect/revoke immediately blocks new work; no inbound firewall configuration; cloud relay downtime preserves local use.

Weeks 7-9: beta and distribution

- Package first desktop platform (macOS or Windows based on pilot user majority); add startup configuration, updater signing and safe diagnostics.
- Recruit 10-20 volunteer developers, including beginners and people with constrained budgets. Use explicit beta terms and privacy notice.
- Instrument opt-in activation events, setup drop-off, 429/503 rates, median time to first response, consent misunderstandings and last-seen latency.
- Perform focused security testing and abuse simulation; clear all P0 vulnerabilities before widening beta.
- Publish installation guide, free/paid/plan capability truth table, supported model list and outage limitations.

Gate: 90% of sampled users correctly explain where their prompt goes; zero known cross-user credential leaks; controlled text test success target at least 95% where quotas are available.

Epics and development tickets

ID	Priority	Ticket and acceptance
CORE-01	P0	Strict authenticated loopback server; invalid key never reaches adapter
CORE-02	P0	OpenAI-compatible text request + model list; schema fixture passes
CORE-03	P0	Provider adapter contract and standardized error enum; mocks cover states
CORE-04	P0	No-logs-by-default instrumentation and local secret vault; canary scan clean
A-01	P0 research	Mock/authorized browser adapter and session lifecycle; no bypass mechanisms
A-02	Gate	Written permission record before any live consumer UI adapter is distributed
C-01	P0	Ollama route: installed-model discovery and 10-case test pass **[S13]**
C-02	P0	Official free provider: auth, request, quota 429, privacy warning
ROUTE-01	P0	User-selected route, no paid fallback by default, destination consent
B-01	P1	Eligible official account grant and documented restricted capabilities **[S05]**
WEB-01	P1	Web console onboarding, status, remote disclosures and device settings
RELAY-01	P1	Device pairing + WSS + owner checks; replay/revoke tests pass
RELAY-02	P1	Remote model call stream and offline 503 with live heartbeat
SEC-01	P0	Local web-origin, dashboard BOLA, secret-leak test suite
DOC-01	P1	Compatibility snippets, supported routes and model truthfulness
MCP-01	P2	Simple local `ask_model` tool uses same auth/router, not special bypass **[S14]**

Recommended sequence: `CORE-01/02/03 -> A-01 and C-01 -> C-02 and ROUTE-01 -> WEB-01/RELAY-01/02 -> SEC-01/beta`. Permission gate A-02 runs independently from the engineering chain.

Testing pyramid and acceptance cases

Layer	Representative cases	Pass criterion
Unit	Request validation, model alias mapping, quota classifier, permission flags, scrub logs	Deterministic fixtures and zero secrets in snapshots
Adapter contract	Success, logout, error, 429, timeout, cancel, malformed stream, unsupported tool	Every adapter passes shared suite
Local integration	curl/JS/Python clients, loopback auth, wrong origin, simultaneous prompts	Correct output/error and no unauthorized execution
Relay integration	Wrong owner, expired key, duplicate frame ID, offline device, revoked pair	No cross-tenant access; bounded clean failure
Security	Cookie/file exposure, SSRF, CSRF, CORS, brute force, update tamper, leaked telemetry	Zero unresolved critical/high findings
Reliability	Simulated provider outage, connection flaps, long output, memory/resource pressure	No unbounded queues or crashes in target tests
UX	Fresh machine install, expired login, quota warning, provider fallback consent	Users can recover without developer support

Concrete scripted end-to-end cases: T01 valid local request; T02 invalid API key; T03 model unavailable; T04 unsupported tool-call; T05 provider 429; T06 browser login expired; T07 canceled streaming client; T08 unauthorized browser origin; T09 cloud offline/local still works; T10 remote device offline; T11 cross-account device ID; T12 pair-code replay; T13 device revocation during stream; T14 secret canary in provider error; T15 paid fallback denied; T16 remote relay privacy setting off; T17 output has unknown exact token count; T18 provider permission flag disabled.

Metrics and SLO candidates

Metric	Pilot target, not observed result	Why it matters
Median time to first local text response	Under 5 minutes	Measures onboarding friction
Eligible local simple-text success	At least 95% within quotas	Distinguishes parsing issues from exhausted allowance
Account separation failures	Zero successes	Critical security property
Gateway P95 overhead versus direct official call	Under 250 ms excluding provider time, for local request	Prevents excessive software-layer latency
Device presence staleness	Dashboard labels offline within 60 seconds after heartbeat loss	Accurate remote expectation
No-content telemetry compliance	100% scripted canary scans pass	Privacy objective
Beta user comprehension	At least 90% identify source and data path	Trust and safe use

Measure baseline first. Avoid collapsing upstream model latency into local bridge overhead, and stratify failures by `quota_exhausted`, `rate_limited`, `permission_disabled`, `login_required` and `provider_unavailable`.

Cost model, sustainability and tradeoffs

No promise of zero total cost: a user's computer consumes electricity, their internet and plan may cost money, providers can change free quotas, and the product's cloud infrastructure has fixed costs. The gateway saves **incremental pay-as-you-go API charges only where an approved account-plan or free/local route actually covers inference**. It cannot make paid API calls free by changing transport. **[S09] [S11] [S31]**

Scenario	Assumed workload	Variable inference cost to product	Other constraint
Local-only Ollama	300 prompts/day on user's PC	Zero proprietary-model API bill	Device hardware/power and model quality
Gemini free tier	User's own quota, eligible region/model	Possibly zero until free allowance exceeded	Provider caps, privacy terms, no 24/7 guarantee
Official account-plan B	Eligible user's permitted plan access	May be included subject to provider conditions	Eligibility, grant constraints and plan limits
Remote relay	100 requests/day with average 2 KB request + 10 KB answer	Low theoretical transfer bytes; cloud hosting remains paid	Device must stay online; concurrency and WSS cost
Paid fallback	User explicitly opts in and sets budget	Charged by upstream provider/account	Must surface projected/actual spend clearly

The traffic numbers above are **planning illustrations, not measured forecasts or vendor quotations**. Do not publish quantitative savings claims until user usage and direct API reference costs are measured in the same model/time/feature settings.

Packaging and distribution concept

Local core: candidate permissive license (Apache-2.0 or MIT), after dependency-license review and business decision. Open sourcing helps trust but does not grant rights to automate third-party UIs. Include a clean adapter SDK and a public policy for approved-provider contributions.

Free hosted dashboard tier: a small number of paired devices and remote requests, designed to subsidize students. Possible sustainable paid tier later: higher remote request ceilings, alerts, teams, device groups and analytics. Never charge for redistributing underlying consumer-plan quota without clear provider approval. Users pay providers separately for any official billed API route.

Initial acquisition: open-source repositories and demos showing a working local Ollama/official-free-API adapter, starter kits for coding agents and student hackathons, clear installation videos, and transparency-led product page. Position as an *account-owned AI router*, not an unlimited proprietary-model API hack.

Founder decisions that cannot be delegated to code

- Which country/region and desktop OS will the beta target first? This affects eligibility, privacy law and packaging.
- Is the initial app open-source, source-available or proprietary? Decide before accepting contributions.
- Which official free provider is the first C route? Select by target users' geography/model and current data policy.
- Is any paid fallback allowed at all in v1? Recommendation: no, unless the user deliberately enables budget limits.
- Are you prepared to pursue formal provider partnerships for A? This determines whether A can become a production feature rather than a research prototype.
- Do you want cloud mode in first public release? Recommendation: prove local stability before activating remote relay.

Final go/no-go criteria

GO for local public alpha if official C path or permitted B path works, local auth/secrets tests pass, advertised capability truth is verified, onboarding is viable, no unauthorized A adapter is exposed and the product explicitly states quotas.

GO for remote beta only after independent cross-tenant/device authorization review, confirmed transit privacy notices, key revocation/cancellation tests, capacity limits and a documented deletion path.

GO for public A connector only with provider authorization covering the exact consumer surface, automation method, user-owned account use, redistribution/commercial distribution, regions and rate policies. If this is denied, do not disguise a prohibited connector as a personal-account feature; prioritize B/C while seeking partnerships.

NO-GO on any critical secret leakage, cross-account request execution, undetected paid fallback, CAPTCHA bypass or a public claim of unlimited proprietary-model access.

Evidence & source register

Sources were checked in October 2026. Their content, access conditions, prices and quotas can change. Project READMEs are evidence of what authors claim to implement, not proof of provider permission or reliability.

[S01] OpenAI consumer Terms of Use (effective January 2026)
https://openai.com/policies/row-terms-of-use/

[S02] Anthropic Consumer Terms of Service (effective October 2025)
https://cdn.sanity.io/files/4zrzovbb/website/43c13f51a6d9755341433f79cf63c966ed9200ea.pdf

[S04] OpenAI Sign in with ChatGPT - Local open-source plan usage
https://developers.openai.com/siwc/token-sharing-open-source

[S05] OpenAI Sign in with ChatGPT - Preview limitations
https://developers.openai.com/siwc/token-sharing-open-source/preview-limitations

[S07] OpenAI Help - Using ChatGPT plans in other apps
https://help.openai.com/en/articles/20001542-using-your-chatgpt-plan-in-other-apps-and-sites

[S09] Google Gemini API billing and free tier
https://ai.google.dev/gemini-api/docs/billing

[S10] Google Gemini API rate limits
https://ai.google.dev/gemini-api/docs/rate-limits

[S11] GroqCloud API rate limits
https://console.groq.com/docs/rate-limits

[S13] Ollama OpenAI-compatible API
https://docs.ollama.com/api/openai-compatibility

[S14] MCP specification release, 28 July 2026
https://blog.modelcontextprotocol.io/posts/2026-07-28/

[S21] Playwright authentication and browser-state guidance
https://playwright.dev/docs/auth

[S31] OpenAI Help - ChatGPT Plus and API separation
https://help.openai.com/en/articles/6950777-what-is-chatgpt-plus

[S39] OpenAI - Self-hosted VM authorized-account notes
https://developers.openai.com/siwc/token-sharing-open-source/self-hosted-vms

> Evidence boundary: this series is a design-and-research package, not a working implementation, legal opinion, measured security certification or promise of free access to any specific proprietary model.